Modules in Python – Complete Detailed Notes

1. What is a Module?

A **module** in Python is a file containing Python code such as:

- functions
- variables
- classes
- statements

saved with the .py extension.

A module helps organize code into separate files so it can be reused in different programs.

Example:

If a file named math_tools.py contains:

```
def add(a, b):
    return a + b
```

Then math_tools.py is a module.

2. Why Use Modules?

Modules are used for:

- Reusability of code
- Better organization
- Easier maintenance
- Breaking large programs into smaller parts
- Sharing code across projects

Without modules, all code stays in one big file.

3. Types of Modules in Python

A. Built-in Modules

Already included with Python.

Examples:

- math
- random
- os
- sys
- datetime
- statistics

B. User-defined Modules

Created by programmer.

Example:

```
# mymodule.py
def greet():
    print("Hello")
```

4. Importing a Module

Use import.

```
import math
print(math.sqrt(25))
```

Output:

5.0

5. How Import Works

When Python sees:

```
import math
```

It:

1. Searches for module
2. Loads module into memory
3. Makes functions available

6. Accessing Module Members

Use dot operator .

```
import math

print(math.pi)
print(math.factorial(5))
```

7. Import Specific Functions

```
from math import sqrt, pi

print(sqrt(16))
print(pi)
```

No need to write math.

8. Import All Functions

```
from math import *
```

Now all functions available directly.

⚠ Not recommended in large projects because names may conflict.

9. Module Alias

Use shorter name with as.

```
import math as m

print(m.sqrt(49))
```

Another example:

```
import numpy as np
```

10. Built-in Important Modules

Module	Use
math	Mathematical functions
random	Random numbers
os	File/folder operations
sys	System operations
datetime	Date and time
statistics	Mean, median

11. Math Module

Used for mathematical calculations.

import math

Common Functions

Function	Work
math.sqrt(x)	Square root
math.pow(a,b)	Power
math.factorial(n)	Factorial
math.ceil(x)	Round up
math.floor(x)	Round down
math.pi	π value

Example:

```
import math
print(math.sqrt(36))
print(math.pi)
```

12. Random Module

Used for random values.

import random

Functions:

Function	Work
randint(a,b)	Random integer
choice(list)	Random item
shuffle(list)	Shuffle list

random()	Float 0 to 1

Example:

```
import random
print(random.randint(1,10))
```

13. OS Module

Used for file and folder operations.

```
import os
```

Functions:

Function	Work
os.getcwd()	Current folder
os.mkdir()	Create folder
os.remove()	Delete file
os.rename()	Rename file
os.listdir()	List files

Example:

```
import os
print(os.getcwd())
```

14. Sys Module

Used for system-level operations.

```
import sys
print(sys.version)
```

Uses:

- Python version
- command line arguments
- exit program

15. Datetime Module

Used for date and time.

```
import datetime

today = datetime.datetime.now()
print(today)
```

16. Statistics Module

Used for averages.

```
import statistics

nums = [10,20,30]
print(statistics.mean(nums))
```

17. Creating User Defined Module

Step 1: Create file calc.py

```
def add(a,b):
    return a+b

def sub(a,b):
    return a-b
```

Step 2: Use in another file

```
import calc

print(calc.add(5,2))
```

18. __name__ == "__main__"

Used to check whether file runs directly or imported.

```
def greet():
    print("Hello")

if __name__ == "__main__":
    greet()
```

If imported, greet() won’t auto-run.

19. Searching Module Path

```
import sys
print(sys.path)
```

Shows where Python searches modules.

20. Reload Module

```
import importlib
import mymodule

importlib.reload(mymodule)
```

Useful after editing module.

21. Package in Python

A **package** is a folder containing multiple modules.

Example:

```
mypackage/
  add.py
  sub.py
```

Used for large projects.

22. Difference: Module vs Package

Module	Package
Single .py file	Folder of modules
Small unit	Larger structure

23. Common Errors with Modules

ModuleNotFoundError

```
import xyz
```

When module not installed or not found.

NameError

Using function without importing.

24. Installing External Modules

Use pip:

```
pip install numpy
```

Examples:

- numpy
- pandas
- requests
- flask

25. Popular External Modules

Module	Use
numpy	Arrays
pandas	Data analysis
matplotlib	Charts
requests	API calls
flask	Web apps

26. Best Practices

✅ Import only needed functions

✅ Use aliases when useful

✅ Organize code into modules

✅ Use meaningful module names

✅ Avoid from module import *

27. Real Life Uses

- Calculator apps (math)
- Games (random)
- Automation (os)
- Reports (datetime)
- Data science (pandas)
- Web development (flask)

28. Viva Questions

Q1. What is a module?

A Python file containing reusable code.

Q2. Difference between built-in and user-defined module?

Built-in comes with Python.

User-defined is created by programmer.

Q3. Why use alias?

To shorten long names.

Q4. What is package?

Folder containing modules.

29. Short Notes for Exam

- Module = .py file
- import loads module
- math.sqrt() uses function
- as gives alias
- Package = collection of modules
- pip installs external modules

30. Memory Trick

I A U P

- **I**mport
- **A**lias

- **U**se functions
- **P**ackage

31. One-Line Summary

Modules in Python are reusable code files that help organize programs and provide ready-made functions for different tasks.